



KAZAKH-BRITISH
TECHNICAL
UNIVERSITY

JSC «Kazakh-British Technical University»
School of IT and Engineering

APPROVED BY

Dean of SITE

Azamat Inanbayev

2025



SYLLABUS

Discipline: Backend Framework. Django

Course code - INFT3131

Term: Fall 2025

Instructors full name: Maratuly Temirbolat

Personal Information about the Instructor	Time and place of classes		Contact information
	Lessons	Office Hours	e-mail
Maratuly Temirbolat	According to the schedule	According to the schedule	t.maratuly@kbtu.kz

COURSE DURATION: 3 credits, 15 weeks, 45 class hours

GENERAL COURSE AIMS

The course aims to introduce students to modern web application development using the Django framework. Starting from zero knowledge, the course is designed to provide both conceptual understanding and practical coding skills through live coding sessions and project-based learning. The general aims of the course are to:

- 1. Introduce web development fundamentals** — including the client-server model, HTTP protocol, and architectural patterns such as MVC/MTV.
- 2. Provide hands-on experience with Django** — guiding students step by step in building full-stack web applications using Django's models, views, templates, and forms.
- 3. Develop proficiency with databases and ORM** — enabling students to design relational models, perform queries, and integrate data into dynamic web pages.
- 4. Familiarize students with user management** — including authentication, authorization, and session handling.
- 5. Introduce basic frontend integration** — using Django Templates with Bootstrap and JavaScript (fetch API) to create responsive and interactive user interfaces.
- 6. Provide an introduction to RESTful APIs** — using Django REST Framework (DRF) to build simple APIs and connect them with frontend code.
- 7. Introduce software engineering practices** — including testing, debugging, and version control with Git.
- 8. Prepare students for deployment** — giving them the skills to deploy a Django project to a production environment (Heroku/Render/Railway).

9. Lay the foundation for Advanced Django — ensuring students are ready to study more complex topics such as CBV, advanced ORM, custom authentication, Docker, Celery, and scalable deployment in the follow-up course.

COURSE DESCRIPTION

The course introduces students to web development using the Django framework. Through live coding and practical projects, students will learn the fundamentals of the MTV architecture, database modeling with Django ORM, templates, forms, and user authentication. The course also includes basic frontend integration with Bootstrap and JavaScript, as well as an introduction to building RESTful APIs using Django REST Framework. By the end of the semester, students will be able to design, implement, and deploy a complete web applications. All the practical and homeworks will be conducted in OquLabs sessions.

COURSE OBJECTIVES

The main objective of this course is to provide students with practical skills in building modern web applications using Django. By the end of the course, students will be able to:

1. Understand the basics of web architecture, HTTP, and the MTV pattern.
2. Set up and configure Django projects and applications.
3. Work with models, migrations, and the ORM to design and query databases.
4. Use templates and forms to create dynamic and interactive web pages.
5. Implement user authentication, authorization, and file handling.
6. Apply Bootstrap and basic JavaScript (fetch API) for frontend integration.
7. Develop simple RESTful APIs with Django REST Framework.
8. Write and run unit tests for Django applications.
9. Deploy a Django project to a production environment.

COURSE OUTCOMES

By the end of this course, students will be able to:

1. Explain the fundamentals of web architecture, HTTP, and Django's MTV pattern.
2. Create and configure Django projects and applications from scratch.
3. Design and manage relational databases using Django ORM and migrations.
4. Build dynamic web pages with templates, forms, and user input validation.
5. Implement user authentication, authorization, and profile management.
6. Integrate static and media files into web applications.
7. Apply Bootstrap and JavaScript (fetch API) for responsive and interactive interfaces.
8. Develop and consume simple RESTful APIs using Django REST Framework.
9. Write basic automated tests to ensure application functionality.
10. Deploy a Django project to an online hosting platform.

COURSE POST REQUISITES:

Web application using Django and Django REST Framework

LITERATURE

Docs:

1. Django documentation: <https://docs.djangoproject.com/en/5.2>
2. Django Rest Framework documentation: <https://www.django-rest-framework.org/>
3. Python documentation: <https://www.python.org/doc/>

COURSE ASSESSMENT CRITERIA

Assessment occurs continuously throughout the course. The evaluation will be based on the levels of (maximums in %):

Type of activity	Final scores
Attendance /participation	10%
Practice	20%
Midterm project defense	15%
Endterm project defense	15%
Final exam	40%
Total	100%

TASKS POINTS EVALUATION

Week	TSIS	Cost (in points)
2	Practice work 1	2
3	Practice work 2	2
4	Practice work 3	2
5	Practice work 4	2
6	Practice work 5	2
9	Practice work 6	2
10	Practice work 7	2
11	Practice work 8	2
12	Practice work 9	2
13	Practice work 10	2
	Total	20

COURSE CALENDAR

Week	Class work		
	Topic	Lectures	Practice
1	Introduction to Web and Django Lecture: Web basics, HTTP, request/response cycle, Django overview, project setup. Create first Django project ("Hello, Django"). Homework: Simple app (time display / random number generator).	2	1
2	Python recap (PEP8, Typing, Docstrings, comments) + flake8 + ruff Lecture: Discuss python basics for more informed coding. Practice work #1: Some functions with detailed code Homework: Write 2–3 functions in PEP8 style, run linter (flake8/ruff).	2	1

3	Git & GitHub Basics Lecture: What is version control? Git basics (init, status, add, commit, log). Branching and merging. Remote repositories (GitHub). Practice work #2: Initialize git repo, make first commits, create a branch, merge changes. Push project to GitHub. Homework: Create a personal repo for the upcoming Django project. Commit and push weekly progress.	2	1
4	Django Project Structure Lecture: MTV architecture, apps, urls, views. Practice work #3: Create new app with basic views and URL mapping. Homework: Build a simple calculator or quote generator.	2	1
5	Templates and Frontend Basics Lecture: Django templates, inheritance, filters. Practice work #4: Connect Bootstrap 5 for styling. Homework: Style pages with Bootstrap (cards, forms).	2	1
6	Models fundamentals Lecture: Models, migrations, queries, relationships. Practice work #5: Add Comment model to Blog. Construct ER diagram. Homework: Write queries using ORM (filter, get, update).	2	1
7-8	Midterm Exam project presentations	2	1
9	Forms and CRUD Lecture: Forms, ModelForms, validation, CRUD operations. Practice work #6: Build a Blog with create/edit/delete posts. Homework: Extend Blog with Bootstrap-styled forms.	2	1
10	Admin and ORM Lecture: Django Admin, customization and ORM (filter, all, select_related, prefetch_related). Practice work #7: Customize Admin for Blog and Comments. Homework: Add search and filters in Admin.		
11	Middleware and Signals Lecture: Middleware basics, Django signals. Practice work #8: Create middleware for request logging, signal on new post. Homework: Extend middleware with custom logic.	2	1
12	Testing in Django Lecture: Django TestCase, pytest basics. Practice work #9: Write tests for Blog models and views. Homework: Add tests for authentication.	2	1
13	Project deployment Lecture: Git, virtual environments, deployment on a server. Practice work #10: Deploy Blog project online. Homework: Share deployed project link.	2	1
14-15	Endterm projects presentations	2	1
16-17	Final Exam		

COURSE ASSESSMENT SCHEDULE

#	Assessment criteria	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	%
1	Attendance / participation	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	10
2	Practice		*	*	*	*	*			*	*	*	*	*			20
3	Midterm							*									15
4	Endterm																15
6	Final exam														*		40
	Total																100

Class sessions – will be a mixture of information by presentations, discussions and practical live coding.

Practice – will reinforce the students' knowledge by practical application of lectured materials.

In-class assessment – will prepare students for their mid-term and final assessment and identify the competence level they have achieved on a related subject matter, the aim being to diagnose potential discrepancies in students' understanding and performance in order to make specific adjustments to the course content and procedures and/or to assign additional assignments to certain individuals or the whole group.

Home assignments – will consolidate the concepts and materials taken during in-class activities, help students to expand the content through diverse background resources and/or practise certain skill areas; they will also develop the students' ability to work individually in exploring and examining related issues.

Midterm projects – students form a group of people where they collaborate all together and formulate projects parts, construct database and distribute the roles which will be their final project.

Endterm projects – students present their projects with 80% of implemented work of final.

Final examination – finished MVP project with tech specification and presentation of functionality.

Grading policy:

Intermediate attestations (on 8th and 15th week) join topics of all lectures, practice, laboratories, SIS, TSIS and materials for reading discussed to the time of attestation. Maximum number of points within attendance, activity, and practices for each attestation is 30 points.

Final exam joins and generalizes all course materials, is conducted in the form of projects. Maximum number of points is 40. At the end of the semester you receive overall total grade (summarized index of your work during semester) according to conventional KBTU grade scale.

ATTENTION!

- 1) If student missed more than **30% of lessons** student receives «**F (Fail)**» grade;
- 2) If for two attestations student receives 29.4 or less points, this student is not accepted to final exam and for all course he (she) receives «**F (Fail)**» grade;
- 3) If student receives 9.4 or less points on final exam, then independently on how many points he (she) received for two attestations, in whole he (she) receives «**F (Fail)**» grade;

In the case of missing or being late for final exam without plausible reason, independently on how many points he (she) received for two attestations, in whole he (she) receives «**F (Fail)**» grade.

- 4) In case of non-compliance of the course project with the **given assignment**, the student is not allowed to **defend the course** project and automatically receives an «**F (Fail)**» grade.

- 5) Practical work must be uploaded to the Teams assignment in written form. If a student fails to upload the work before the specified deadline, the work will not be considered. A score of zero is given.

- 6) All written work must be completed according to the **standard** of KBTU. The standard and template are attached to the TEAMS platform. The **Teams team** has added a video lesson on preparing reports according to standards.

- 7) All text materials are available on the <https://wsp.kbtu.kz/> and TEAMS platform.
- 8) If a student does not pass the **practice task** on time, then admission to the next **practice** will be only after the mandatory completion of the previous work. Points for previous work will be lost, but feedback will be given on correct completion.
- 9) To qualify for the exam, each student must individually pass and complete **all practical works**.
- 10) If plagiarism is detected in the text of the work, in the program code, or if the work does not meet the standards, the work is canceled.

ACADEMIC POLICY:

1. Cheating, duplication, falsification of data, plagiarism are not permitted under any circumstances!
2. Students must participate fully in every class. While attendance is crucial, merely being in class does not constitute "participation". Participation means reading the assigned materials, coming to class prepared to ask questions and engage in discussion.
3. Students are expected to take an active role in learning (the instructor will provide the information and guidelines to do this).
4. Students must come to class on time.
5. Students are to take responsibility for making up any work missed.
6. Make up tests in case of absence will not normally be allowed.
7. Mobile phones must always be switched off in class.
8. Students should always show tolerance, consideration and mutual support towards other students.

RULES FOR ATTENDING PRACTICAL CLASSES:

1. Each student must read the safety instructions. Sign the safety register.
2. It is forbidden to start the laboratory equipment (switch on the power supply) and its individual parts without the permission of the teacher.
3. It is forbidden to swap equipment components.
4. It is prohibited to take equipment components out of the classroom.
5. **The student is responsible for the integrity of the equipment during work. If a malfunction is detected, the student must immediately turn off the power and inform the teacher about the failure.**
6. Students should use connections carefully. Avoid breaking contacts and connections.
7. No more than two people are allowed per workstation.
8. The maximum number of students in a practical lesson cannot exceed 18 (9 stations available).
9. Students who are late for a practical lesson are not allowed into class due to safety precautions.

Grade		Achievement percentage	Assessment criterion
«Excellent»	A	95-100%	This grade is given when the student: demonstrated a complete understanding of the course material; did not make any errors or inaccuracies;

	A -	90-94%	completed control and laboratory work in a timely and correct manner, and submitted reports on them; demonstrated original thinking; submitted control quizzes on time and without any errors; completed homework assignments; engaged in research work; independently used additional scientific literature in studying the discipline; was able to independently systematize the course material.
«Good»	B+	85-89%	This grade is given when the student: Has mastered the course material at no less than 75%; Did not make gross errors in responses; Timely completed control and laboratory work and submitted them without fundamental remarks; Correctly completed and timely submitted control tests and homework assignments without fundamental remarks; Utilized additional literature as indicated by the instructor; Engaged in research work, made non-fundamental errors, and fundamental errors corrected by the student themselves; Managed to systematize the course material with the help of the instructor.
	B	80-84%	
	B-	75-79%	
	C+	70-74%	
«Satisfactory»	C	65-69%	This grade is given when the student: Has mastered the course material no less than 50%; Required assistance from the instructor when completing control and laboratory work, homework assignments; Made inaccuracies and non-fundamental errors when submitting control tests; Did not demonstrate activity in research work, relied solely on the educational literature indicated by the instructor; Experienced more difficulty in systematizing the material.
	C-	60-64%	
	D+	55-59%	
	D	50-54%	

Instructor

Maratuly Temirbolat

Minutes #1 of School of Information Technology and Engineering meeting on August 18, 2025